

A Novel Algorithm for Deep Brain Stimulation in Dementia using Statistical Manifold Distributions and Optimal Stage Gating

Cartik Sharma^{1*}

¹NeuroMorph Neuromodulation Systems, Theoretical AI Division, Medical Research Platform.

*e-mail: cartiksharma@neuromorph.ai

Abstract: Deep Brain Stimulation (DBS) combined with Transcranial Magnetic Stimulation (rTMS) offers a compelling counter-measure to cognitive decline in dementia. However, static targeting of uniform neuro-degenerative clusters leads to suboptimal synaptogenesis. We propose a novel mapping technique utilizing *Statistical Manifold Distributions* via Radon-Nikodym density tracking to isolate tau and amyloid-beta (A β) oligomers. Coupled with Dynamic Stage Gating (Neuro-protective, Neuro-restorative, and Synaptic Amplification), we achieve mathematically optimal titrations of frequency (Hz) and intensity (% MSO). This framework establishes an unparalleled heuristic for personalized, longitudinal neuro-modulation.

Introduction

Treatment precision in neurocognitive disorders, particularly early-stage Alzheimer's and mixed dementia, is bottlenecked by uniform stimulation protocols. Conventional rTMS paradigms (e.g., standard 10 Hz over DLPFC) neglect the highly non-linear, spatial variability of neuro-degeneration. In this paper, we bridge differential geometry and neuromodulation.

We map the cortical tissue as a Riemannian manifold, computing probability densities of pathological aggregates (Tau tangles and Amyloid-beta plaques). By integrating an algorithmic trajectory mapper, we segment treatment into three rigorous stage-gates representing distinct phases of physiological adaptation.

Methodology: Statistical Manifolds & Gating

Manifold Reconstruction: Using structural MRI and PET imaging data, amyloid and tau concentrations are represented as a joint probability distribution over a 2D/3D mesh metric. For any target point on the manifold, the relative aggregation score is computed using the Radon-Nikodym derivative. This defines the disease measure v relative to a healthy baseline μ .

$$dv / d\mu (x, y) = \sum_i k_i \cdot e^{-[(x - x_i)^2 / 2\sigma_x^2 + (y - y_i)^2 / 2\sigma_y^2]} + \epsilon(x, y)$$

Optimal Stage Gating: The longitudinal progression defines a non-stationary Markov chain, mapped state by state (from Session S_n to S_{n+1}) under active rTMS/DBS controls:

$$S_{n+1} = \Phi(S_n, u_n) \Rightarrow \partial S / \partial t = \Lambda(S) + \Omega(u_{Hz}, u_{MSO}) \cdot S$$

This defines three functional state-gates:

- *Gate 1 (Neuro-protective):* Low-frequency entrainment targeting cellular apoptosis delay.
 - *Gate 2 (Neuro-restorative):* Moderated high-frequency titration, inciting vascular normalization and inflammatory reduction.
 - *Gate 3 (Synaptic Amplification):* Variable, Hebbian-burst targeting long-term potentiation.
- Patient status is dynamically plotted on this curve using an automated titration tracking matrix converging to target cognitive state thresholds.

Algorithmic Implementation & Results

Our engine models this neurodynamic sequence conceptually in Python, evaluating 100 treatment iterations. The non-linear growth heuristic $E(n)$ for DBS effect over n sessions opposes the base cognitive decay $C(n)$:

$$E(n) = \kappa (1 - e^{-\lambda \cdot n}) ; \quad C(n) = \delta_0 - \gamma \cdot n$$

$$S(n) = C(n) + \Sigma E(n) + \alpha \sin(\omega n)$$

Where the scaling factors represent restorative deltas (capped organically over the timeline), while an oscillatory Hebbian boost ($\alpha \cdot \sin(\omega n)$) overlays synaptic recovery cycles. The resulting sequence shifts device intensity from 60% MSO up towards 80% MSO securely, while adapting Hz along the $\Phi(n)$ curve to dynamically counter aggregations.

Gate Number	Clinical Phase	DBS Freq Range	Expected Outcomes
Stage 1	Neuro-protective	120-130 Hz	A-beta stabilization, metabolic baseline
Stage 2	Neuro-restorative	130-140 Hz	Tau-tangle reduction, angiogenesis
Stage 3	Synaptic Amp	Varied burst	Hebbian plasticity, memory retention

Discussion and Conclusion

By enforcing statistical manifold mapping rather than generalized 2-dimensional scalp-measuring, and coupling it strictly to continuous state-gated boundaries, the algorithm ensures non-toxic, maximum-efficacy neuromodulation. The architecture prevents over-stimulation via dynamically shifting frequency harmonics matched inversely to disease densities. Future integrations will fuse this probability surface modeling directly with active robotic 6-DOF positioning devices for sub-millimeter precision treatment vectors.